

9 Financial Analysis Technique

Activity Ratio	
• Inventory turnover=$\text{COGS}/\text{average inventory}$ • Days of inventory on hand=$365/\text{inventory turnover}$	反映企业经营效率，以及企业的各种资源是否得到了充分利用。这里的资源即包含了流动资产和流动负债，还包括长期资产。
• Receivable turnover=$\text{revenue}/\text{average receivable}$ • Days of sales outstanding=$365/\text{receivable turnover}$	
Payable turnover = $\text{purchase}/\text{average payable}$ Number of days payables= $365/\text{payable turnover}$	
• Total assets turnover=$\text{revenue}/\text{average assets}$ • Fixed assets turnover=$\text{revenue}/\text{average fixed assets}$ • Working capital turnover=$\text{revenue}/\text{average working capital}$	

DOH = days of inventory on hands

DSO = days of sales outstanding

Cash conversion cycle = DOH + DSO – Days of payables

Liquidity Ratio	
• Current ratio=$\text{current assets}/\text{current liabilities}$ • Quick ratio=$(\text{cash}+\text{short-term marketable security}+\text{receivable})/\text{current liability}$ • Cash ratio=$(\text{cash}+\text{short-term marketable security})/\text{current liability}$	反映企业履行其长期负债义务的能力。
• Operating cycle=$\text{days of inventory on hand}+\text{days of sales outstandings}$ • Cash conversion cycle=$\text{operating cycle}-\text{number of days of payable}$	

Solvency Ratio	
• debt-to-assets ratio=$\text{total debt}/\text{total assets}$ • debt-to-capital ratio=$\text{total debt}/(\text{total debt}+\text{total equity})$ • debt-to-equity ratio=$\text{total debt}/\text{total equity}$ • financial leverage=$\text{average total assets}/\text{average total equity}$	反映企业履行其长期负债义务的能力。
• interest coverage ratio=$\text{EBIT}/\text{interest payments}$ • fixed charge coverage ratio= $(\text{EBIT}+\text{lease payments}) / (\text{interest payments}+\text{lease payments})$	

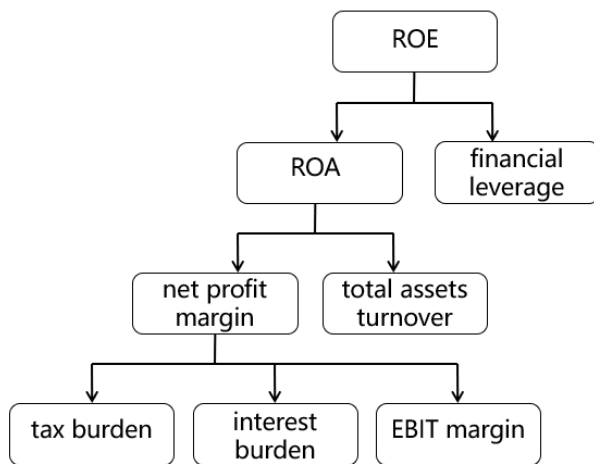
Profitability Ratio

- **gross profit margin**=gross profit/revenue
- **operating profit margin**=operating profit/revenue
- **pre-tax margin**=earning before tax/revenue
- **net profit margin**=net income/revenue

- **return on assets (ROA)** =net income/total average assets
- **return on capital**=EBIT/ (total average debt+ average equity)
- **return on equity (ROE)** =net income/total average equity

反映企业赚取利润的能力。

Dupont Analysis — 5 Steps



$$\begin{aligned}
 \text{ROE} &= \frac{\text{NI}}{\text{Equity}} \\
 &= \frac{\text{NI}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Equity}} \\
 &= \frac{\text{NI}}{\text{Revenue}} \times \frac{\text{Revenue}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Equity}} \\
 &= \frac{\text{NI}}{\text{EBT}} \times \frac{\text{EBT}}{\text{EBIT}} \times \frac{\text{EBIT}}{\text{Revenue}} \times \frac{\text{revenue}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Equity}}
 \end{aligned}$$